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Evaluating the Deployment Utility of a Frozen Policy-Update Procedure: Scale, Completion Preference, and Joint Evidence

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Abstract

Policy updates selected by proxy scores need not yield positive operational utility when the complete selection and adoption procedure is deployed. We define procedure-level deployment utility as expected incumbent-relative utility under stated operating conditions, including incumbent-retaining decisions. We characterize three elementary properties: reversal frequency alone does not determine mean procedure utility; fixed-selection revaluation under affine completion utility yields a linear preference frontier; and repeated deployments reduce within-root noise without increasing the number of independent roots. These characterizations introduce no new fidelity metric or inference algorithm. We instantiate the assessment for a frozen execution-policy procedure in a synthetic continuous double auction. Fifty independent roots each generate a shared pool of 16 candidates, followed by one proxy-based selection per root–regime pair (100 selection decisions) and paired evaluations on separate interactive books across two liquidity regimes, three order sizes, and two inventory-response settings. Under the prespecified 100-basis-point completion preference and approximate simultaneous bands for the original 24-estimate family, all four largest-scale means are negative, while lower-scale means and response and interaction contrasts remain unresolved. Fixed-selection revaluation over four preferences shows that the deployment conclusion depends on utility and inferential-family choice; at 100 basis points, the same liquid-regime estimates include zero in the wider 64-estimate diagnostic family. A separate outcome-aware diagnostic at the selection scale estimates positive proxy-minus-deployment gaps of 0.95–1.31 bps across four regime–response cells, with a separate simultaneous band; because it combines winner selection, finite proxy-sample noise, and replay versus interactive worlds, it is not a causal estimate of proxy bias. An exploratory post-outcome fixed-selection diagnostic estimates break-even preferences with uncertainty reported separately. The study provides bounded empirical evidence and an auditable reporting specification. It establishes neither general agent reliability, a universal scaling law, nor real-market execution validity.

1 Introduction

Machine-learning systems increasingly generate candidate policies, models, or programs and use proxy evaluations to choose updates before deployment. The selected candidate's proxy score is not the operational utility of the complete procedure that generated and selected it. Procedure-level utility also depends on whether an update is adopted, the conditions under which the deployed system operates, how outcomes are valued, and the population of independent update decisions to which the estimate is meant to generalize. Retaining the incumbent is one possible output of the procedure, not a missing observation.

Several established directions address parts of this problem. Off-policy evaluation estimates a target policy's value from data generated by another policy (Jiang & Li, 2016; Thomas et al., 2015); safe policy improvement

[1]**Reproducibility.** Code and saved-outcome data are available at `https://github.com/computational-decision-lab/policy-update-deployment-utility`; dated release `202611050855`.

studies how to construct updates with guarantees relative to a baseline (Laroche et al., 2019); and the closest TMLR precedent, Abdelwahed et al. (2024), uses conservative off-policy value estimates to inform deployment of offline-learned policies. Active offline policy selection allocates limited online interaction adaptively across candidate policies (Konyushkova et al., 2021). Agent-evaluation work studies benchmark coverage and the reliability of model, scaffold, and task comparisons (Kapoor et al., 2026; Hardy et al., 2026; Kehkashan et al., 2026). These works motivate careful deployment assessment, but their principal targets differ from ours: we estimate incumbent-relative utility of a complete frozen update-selection procedure, including adoption and no-update decisions, on fresh interactive deployments.

We use *procedure-level deployment utility* for the expected incumbent-relative utility of a frozen update-selection procedure under declared operating conditions and an explicit utility. Its assessment specifies the procedure, adoption rule, independent-root population, deployment conditions, utility, and inferential family. It separates three questions: whether a selected candidate improves on a proxy, whether the complete procedure has positive expected utility, and whether the independent evidence supports a deployment claim. The resulting assessment specification organizes existing design and statistical choices; it is not a new universal metric, selection algorithm, or general theory.

We instantiate this assessment in a synthetic continuous double auction. A frozen replay-proxy procedure selects among a fixed pool of 16 candidates per root, after which selected policies and incumbents are compared on separate interactive books. The principal design varies order scale, liquidity regime, and inventory-response condition over a shared population of 50 independent generation roots. It is a controlled evaluation substrate, not a calibrated professional market. The candidate-pool size is fixed, so this paper does not estimate how search pressure changes with the number of candidates.

Two research questions organize the evidence. **RQ1:** How does operating scale affect the expected utility of the frozen update-selection procedure under a prespecified utility? **RQ2:** How does fixed-selection revaluation across completion preferences change estimated utility, and how does inferential-family choice change the evidence supporting a deployment claim? The original market analysis addresses RQ1. Utility revaluation and the break-even analysis address RQ2 as post-outcome diagnostics. Root deletion and alternative bands are sensitivity audits of the same cohort, not independent confirmation.

The paper makes three bounded contributions. First, it states the procedure-level estimand and gives three elementary decision-theoretic and sampling properties that clarify magnitude dependence, utility frontiers, and the role of independent roots; these are formal characterizations, not a new estimator or nontrivial general theorem. Second, it reports a separately frozen interactive-market cohort with 50 shared independent roots, 100 selection decisions, 2,400 paired deployment records, and a prespecified 24-estimate inferential family. Third, it provides an auditable reporting specification that binds the procedure, adoption rule, population, operating conditions, utility, and evidence family, alongside reproducible saved-outcome reanalysis.

Under the original 100 bps completion preference and approximate simultaneous bands for the 24-estimate family, the four largest-scale procedure means are negative. Lower-scale cell means and response and interaction contrasts remain unresolved. At 100 bps, the liquid-regime point estimates are unchanged but their intervals include zero in the wider 64-estimate diagnostic family. A separate post-outcome comparison at the $s = 0.05$ selection scale estimates positive proxy-minus-deployment gaps of 0.95–1.31 bps across four regime–response cells; this auxiliary family does not identify causal proxy bias. This evidence is conditional on one frozen procedure and synthetic market. A restricted planner-code case with only three generation roots is retained as an auxiliary result in the Supplementary Material and does not support broad agent generalization.

2 Related work and contribution boundary

2.1 Policy evaluation and safe improvement

Off-policy evaluation estimates the value of a target policy from logged trajectories generated by another policy. Doubly robust evaluation combines importance weighting and value modeling for this setting (Jiang & Li, 2016); high-confidence off-policy evaluation constructs a lower confidence bound for a policy's expected

return from logged trajectories (Thomas et al., 2015). Safe Policy Improvement with Baseline Bootstrapping (SPIBB) addresses a constructive problem: it uses a baseline policy where evidence is insufficient to seek policy improvement with safety guarantees (Laroche et al., 2019). The closest TMLR precedent is Abdelwahed et al. (2024), which uses conservative off-policy policy-value evaluation and confidence bounds to support a deployment decision for offline-learned policies. Our target is different: rather than evaluating only a fixed learned policy or proposing a safe-improvement algorithm, we estimate the expected incumbent-relative utility of the complete frozen generation, proxy-selection, adoption, and incumbent-retention procedure. The present study uses separate interactive deployment evaluations; it does not propose an off-policy estimator or a safe-improvement guarantee.

2.2 Policy selection under limited online evaluation

Konyushkova et al. (2021) combine logged-data off-policy estimates with a limited online-interaction budget, then use Bayesian optimization to choose which candidate policy to evaluate next. This is an adaptive allocation problem: the evaluation sequence changes in response to earlier observations. We instead freeze the 16-candidate proxy-selection and adoption procedure, evaluate its resulting choices on fresh paired deployments, and estimate procedure-level utility across operating scales and completion preferences. We do not optimize the interaction budget or compare adaptive evaluation strategies.

2.3 Stable policy learning and sampling risk

Barnhard et al. (2026) study how algorithmic stability shapes the tradeoff between expected welfare and sampling risk when a policy is learned from an experimental sample and then implemented at scale. They propose policy-vote bagging and derive bounds linking estimation accuracy, subsample size, and welfare variation, including an exact guarantee under CARA utility. Their target is the stability and welfare risk of the learning algorithm under sample perturbations. Our target is different: we freeze candidate generation, proxy selection, adoption, and incumbent retention, then evaluate incumbent-relative utility on fresh interactive deployments across declared conditions. We do not measure sampling risk in the Stable Policy Learning sense, and we keep this confirmatory assessment distinct from post-outcome diagnostics.

2.4 Agent evaluation and measurement reliability

The Holistic Agent Leaderboard (HAL) develops evaluation infrastructure and compares models, scaffolds, and benchmarks across 21,730 rollouts (Kapoor et al., 2026). Hardy et al. (2026) decompose variation in agent rankings across models, scaffolds, and tasks and show that evaluation reliability depends on the intended ranking claim. A broader review reports gaps between task-success metrics and deployment-relevant dimensions in the benchmarks it surveys (Kehkashan et al., 2026). This work primarily assesses agent capabilities, benchmark infrastructure, or ranking reliability. We instead target the expected incumbent-relative utility of a procedure that generates and selects updates; the independent population is update-generation roots under a specified frozen design, not a population of models or tasks. The completion preference in our market utility is not a measure of API cost, compute efficiency, or safety.

2.5 Proxy optimization and selection bias

Optimizing an imperfect reward proxy can increase its discrepancy from a reference reward, while reward-model ensembles address this effect in the studied reinforcement-learning settings (Gao et al., 2022; Coste et al., 2024). Model-selection bias and adaptive leaderboard submissions provide related accounts of evaluation reuse and selection effects (Cawley & Talbot, 2010; Blum & Hardt, 2015). These works motivate separating proxy selection from subsequent deployment evaluation. Our main cohort fixes the candidate pool at 16 and does not compare pool sizes, so it estimates no search-pressure effect or search-scaling law. Utility sensitivity also holds the selected policies fixed; it does not assess re-selection under alternative completion preferences.

Aspect	Related PIVOT work	This study
Research target	Fidelity of proposed transitions and allocation of validation evidence	Deployment utility of a fixed selection procedure under scale and completion preferences
Method role	Fidelity quantities and PIVOT-KG	Inherited quantities and standard resampling; no new allocator
Market material	Fixed updates on historical price paths with formula-based virtual fills and impact	Replay-selected candidates on event-driven matching-engine books; 50 shared roots
Empirical claims	Game/response and validation-allocation results, including driving stress tests	Original 24-estimate deployment family; separate utility and root diagnostics
Auxiliary evidence	Prior manuscript's own experiments	Distinct three-root planner-code study, reported only in the supplement

Table 1: Scope distinction based on the related manuscript and its submitted supplementary archive. Shared motivation and inherited methods are acknowledged; the current experiments are not offered as evidence for a new PIVOT algorithm.

2.6 Policy-dependent worlds and market simulation

Performative prediction studies settings in which model deployment changes the distribution used to measure performance (Perdomo et al., 2020); WILDS organizes evaluation under specified distribution shifts (Koh et al., 2021). In financial simulation, recent work studies calibration, endogenous order flow, and the response of generated limit order books to injected orders (Noble et al., 2026; Nigmatulin et al., 2026). These works examine related but distinct objects: distributional response, robustness, or simulator realism. Our estimate is conditional on one controlled synthetic world and does not identify real-market execution impact or certify simulator fidelity.

2.7 Relationship to prior Improvement Fidelity work

Anonymous (2026) define Improvement Fidelity and transition-level summaries, study operator and world-response effects, and propose PIVOT-KG to allocate high-fidelity queries according to decision relevance. Their supplementary code also contains virtual execution/participation examples. In the PIVOT E6 financial fixtures, a specified position-size change is assessed on supplied historical price paths using replay and formula-based virtual fills with participation/impact parameters. Here, one frozen replay-selection procedure chooses among 16 candidates per root and is evaluated on event-driven continuous-double-auction books. The cohorts, outcome files, execution mechanisms, and inferential targets differ; the earlier financial results are not reused as current evidence. We inherit and attribute relevant evaluation concepts and standard paired/bootstrap operations, but add no new fidelity metric, validation allocator, or transition-level theorem. The present contribution is procedure-level deployment utility under operating conditions and completion preferences. Table 1 states the substantive boundary. An anonymous copy of the actual related conference manuscript is supplied for comparison.

The exchange uses a pinned, unmodified PythonMatchingEngine implementation (Merlos, 2019). Its engine lineage differs from ABIDES (Byrd et al., 2019; Amrouni et al., 2021). The surrounding policies, generator, and completion utility intentionally retain compatibility with our earlier development implementation. Thus, engine independence does not mean that every modeling choice was developed independently. Those development results are not part of this paper's empirical evidence.

Market-simulation research addresses a complementary validation problem. Noble et al. (2026) calibrate queue-reactive state and event-time distributions from market data and introduce trade-flow feedback for impact and reversion. The LOB-Impact Benchmark injects metaorders into five autoregressive generators across five NASDAQ stocks to assess own-order response and subsequent relaxation (Nigmatulin et al., 2026). These studies examine whether simulated execution dynamics are credible. We instead estimate the utility of the procedure producing a deployed policy, conditional on a specified synthetic world. The market is an evaluation environment, not a claimed new or calibrated market-impact model.

3 Formal target and characterizations

3.1 Frozen procedures, adoption, and utility

Let R index an independent generation and evaluation root. A fixed procedure $\mathcal{P}$ generates candidates, evaluates them using pre-deployment information H_R, selects a provisional candidate $\widetilde{\pi}_R$, and decides whether to replace an incumbent $\pi_{0,R}$. Write the actual adoption event as $A_R = a(H_R)$ and the deployed policy as $\widehat{\pi}_R = \widetilde{\pi}_R$ on A_R, otherwise $\widehat{\pi}_R = \pi_{0,R}$. The adoption gate is frozen using pre-deployment information and is not conditioned on the deployment outcomes used to estimate reversal risk.

For deployment condition z and utility U_p, define a root's conditional mean gain

$$g_R(z;p) = \mathbb{E}_\omega \left[U_p\{\widehat{\pi}_R, W_z(\widehat{\pi}_R, \omega)\} - U_p\{\pi_{0,R}, W_z(\pi_{0,R}, \omega)\} \mid H_R \right], \tag{1}$$

where W_z denotes the policy-dependent deployment world and ω its fresh exogenous randomness. A common exogenous stream couples the two branches; each branch has its own evolving book. Matching streams does not force endogenous paths or fills to agree. If the incumbent is retained, the gain is exactly zero.

Our primary estimand is the procedure mean utility

$$\mu_{\mathcal{P}}(z;p) = \mathbb{E}_R[g_R(z;p)]. \tag{2}$$

The expectation includes the randomness of candidate generation and selection under the declared procedure. It excludes changes to that procedure, simulator misspecification, and research costs unless those costs are part of U_p. The market utility here does not charge for generating or validating candidates. The experiment therefore evaluates deployment value, not the net profitability of conducting the research programme.

An adoption-conditioned reversal probability, $\mathbb{P}\{g_R(z;p) < 0 \mid A_R\}$, is a complementary inherited risk description (Anonymous, 2026). It is undefined when adoption probability is zero. No-update outcomes remain in μ, and

$$\mu_{\mathcal{P}}(z;p) = \mathbb{P}(A_R)\mathbb{E}_R[g_R(z;p) \mid A_R] \tag{3}$$

when $\mathbb{P}(A_R) > 0$.

Formal characterization 1 (Magnitude dependence). Let $q = \mathbb{P}(A_R) > 0$, let $G_R = g_R(z;p)$ on A_R, and define $G_+ = \max(G_R, 0)$ and $G_- = \max(-G_R, 0)$. Then

$$\mu_{\mathcal{P}}(z;p) = q\left\{\mathbb{E}[G_+ \mid A_R] - \mathbb{E}[G_- \mid A_R]\right\}. \tag{4}$$

Consequently, adoption and reversal frequencies alone do not determine procedure utility; the magnitudes of adopted gains and losses are also required. *Proof.* Condition on A_R in Equation 3 and write $G_R = G_+ - G_-$. For example, with $q = 1/2$, adopted gains $+2$ and -1 give mean 0.25, whereas replacing the loss with -10 gives mean -2; both have conditional reversal frequency $1/2$. □

In the market data, each root gain is estimated using four deployment repetitions. Reported reversal counts use the sign of that estimated root mean, with the recorded 10^{-9} bps numerical tolerance. They are descriptive observed-root sign counts, not noise-corrected estimates of the latent conditional probability above, and have no simultaneous risk guarantee.

3.2 Deployment-utility assessment specification

The report can be indexed by $\mathcal{D} = (\mathcal{P}, A, \mathcal{R}, z, U_p, F)$, where $R \sim \mathcal{R}$ identifies the independent root population and F the family of claims assessed jointly. This notation collects the design choices underlying Equation 2; it introduces no new estimand. Table 2 maps the specification to the principal experiment. In particular, 50 is the number of sampled roots, not the definition of the target population.

A prospective application freezes candidate generation, selection, and adoption; declares deployment conditions, utility, and the root population; and fixes the inferential family before examining outcomes. It then estimates paired gains, retains incumbent decisions, and reports joint evidence with completeness and sensitivity audits. Any outcome-aware revision receives a separate diagnostic identity. This workflow organizes standard evaluation operations, rather than establishing their validity in an untested domain.

Element	Question to specify	Principal experiment
Procedure $\mathcal{P}$	How are updates produced and chosen?	Frozen generator, 16 candidates, replay-proxy selection
Adoption A	When is the incumbent replaced?	Proxy gain exceeds 10^{-9} bps; otherwise retain it
Population $\mathcal{R}$	Which independent decisions are represented?	Roots from the frozen generator/world design; 50 sampled roots shared across regimes
World W_z	Where will the decisions operate?	Two liquidity regimes, three order sizes, two response settings
Utility U_p	How are operational outcomes valued?	Shortfall and unfinished fraction; primary $p = 100$ bps
Estimand $\mu_{\mathcal{P}}$	What average utility is sought?	Incumbent-relative gain of the whole procedure, including no-update zeros
Evidence F	Which claims share uncertainty control?	Original 24 estimates; 20,000 whole-root bootstrap draws

Table 2: Deployment-utility assessment specification. The general questions can be reused with other procedures and utilities; their empirical validation here is confined to the stated studies.

Stage	Frozen assessment procedure
Input	Freeze candidate generation, proxy selection, adoption rule, incumbent, target root population, deployment conditions, utility, and inferential family.
1	For each independent root, generate candidates and select a provisional update using only pre-deployment proxy information.
2	Apply the frozen adoption rule; record an incumbent-retaining decision as a zero procedure gain.
3	For each declared deployment condition, evaluate selected and incumbent policies on fresh paired deployments, keeping the complete root vector.
4	Average deployment repetitions within root; estimate the procedure mean and construct joint evidence by resampling whole roots across the declared family.
Output	Report estimates, uncertainty, completeness, sensitivity analyses, and the scope of any deployment interpretation.

Table 3: Operational steps for a procedure-level deployment assessment. The statistical operations are standard; the contribution is an explicit reporting target and its bounded empirical application.

3.3 Decision-theoretic and sampling properties

For the affine completion utility used in the market, define

$$U_p = -S - pf, \qquad a_z = \mathbb{E}_R[S_{\text{incumbent}} - S_{\text{selected}}], \qquad b_z = \mathbb{E}_R[f_{\text{incumbent}} - f_{\text{selected}}], \tag{5}$$

where expectations include fresh deployment randomness and no-update roots. Fixed selections therefore satisfy

$$\mu_{\mathcal{P}}(z; p) = a_z + pb_z. \tag{6}$$

Formal characterization 2 (Fixed-selection utility frontier). Under Equation 5, the procedure mean is affine in p. If $b_z \neq 0$, its mathematical zero is $p_z^* = -a_z/b_z$. Thus changing p can change the assessed utility without changing any selected policy or deployment outcome; it does not evaluate re-selection under the new preference. *Proof.* Subtract the incumbent and selected utilities and take expectations: $\mathbb{E}[-S_{\text{selected}} - pf_{\text{selected}} + S_{\text{incumbent}} + pf_{\text{incumbent}}] = a_z + pb_z$. Solving $a_z + pb_z = 0$ gives the stated zero. □

For inference, let $C_z \subseteq \mathbb{R}^2$ be a joint confidence set for (a_z, b_z). The corresponding projected zero set is

$$\mathcal{P}_z^*(C_z) = \{p \geq 0 : \exists (a, b) \in C_z,\ a + pb = 0\}. \tag{7}$$

A conservative sign classification at preference p can use

$$\underline{\mu}_z(p) = \inf_{(a,b)\in C_z} (a + pb), \qquad \overline{\mu}_z(p) = \sup_{(a,b)\in C_z} (a + pb). \tag{8}$$

The set in Equation 7 is reported only for the separate outcome-aware break-even diagnostic in Section D; it is not part of the original 24-estimate family. If $\underline{\mu}_z(p) > 0$, the confidence set supports a positive mean; if $\overline{\mu}_z(p) < 0$, it supports a negative mean; otherwise the sign remains unresolved, conditional on the stated coverage event.

3.4 Joint evidence across deployment conditions

For regime g, response ρ, and scale s, average the four paired repetitions within root r:

$$D_{r,g,\rho,s} = \frac{1}{4} \sum_{k=1}^{4} \left(U^{\text{selected}}_{r,g,\rho,s,k} - U^{\text{incumbent}}_{r,g,\rho,s,k} \right). \tag{9}$$

Each root contributes its complete vector of cell gains and contrasts. Both regimes use the same 50 roots, so the independent sample size is 50, not the 100 root–regime decisions or the 2,400 paired repetitions. The population interpretation assumes independently generated roots from the frozen design. Shared regimes and repeated deployments improve within-root comparisons without adding independent roots.

The original inferential family contains 12 cell means, four largest-minus-smallest scale contrasts, six response contrasts, and two scale-by-response interactions. For root vector $X_r \in \mathbb{R}^{24}$, let $\overline{X}$ be its sample mean. Each of 20,000 bootstrap draws resamples 50 whole root vectors with replacement and computes

$$T_b = \max_{1\le j\le 24} |\overline{X}^{(b)}_j - \overline{X}_j|. \tag{10}$$

The empirical 95th percentile q gives the bands $[\overline{X}_j - q, \overline{X}_j + q]$. This is the prespecified unstandardized maximum-deviation construction, with approximate simultaneous coverage conditional on the experiment. It is not an exact small-sample guarantee. All 24 estimands are retained even when the main text highlights only a subset.

Formal characterization 3 (Independent-root target and variance decomposition). For a fixed deployment condition, suppose $D_{R,1}, \ldots, D_{R,m}$ are conditionally independent paired deployment gains given the frozen pre-deployment information H_R, with conditional mean g_R and variance σ^2_R. Define $\widehat{g}_R = m^{-1} \sum_k D_{R,k}$ and $\widehat{\mu}_{\mathcal{P}} = n^{-1} \sum_{r=1}^{n} \widehat{g}_r$ over independent roots. Then

$$\mathbb{E}[\widehat{\mu}_{\mathcal{P}}] = \mu_{\mathcal{P}}, \qquad \operatorname{Var}(\widehat{\mu}_{\mathcal{P}}) = \frac{\operatorname{Var}_R(g_R)}{n} + \frac{\mathbb{E}_R[\sigma^2_R]}{nm}. \tag{11}$$

Proof. Conditional independence gives $\mathbb{E}[\widehat{g}_R \mid H_R] = g_R$ and $\operatorname{Var}(\widehat{g}_R \mid H_R) = \sigma^2_R/m$. Apply iterated expectation and the law of total variance to the mean of n independent root-level estimates. □

This decomposition makes the inferential unit explicit: increasing m reduces within-root deployment noise, whereas increasing n reduces both root-population and within-root contributions. The 2,400 paired records in the market study therefore do not constitute 2,400 independent observations; the primary independent unit is the 50-root cluster.

For an ordered scale grid, a positive initial segment requires every included cell's joint lower bound to exceed the numerical tolerance. Its length is zero if the first cell is unresolved. An upper bound above zero merely fails to exclude positive value. Neither summary supplies a continuous capacity threshold, monotonicity between scales, or a guarantee about every adopted update. Constant empirical columns require special care: a degenerate bootstrap is not proof of absent population variation. The common radius in this experiment also applies to the observed constant columns.

4 Empirical instantiation: a frozen procedure in a synthetic market

The experiment instantiates the assessment specification in a controlled execution environment. Its target is one frozen procedure's value at the declared operating conditions. The grid and root population delimit the claim; they do not establish a general law of agent deployment or market capacity.

4.1 World, policies, and selection

The protocol was frozen on September 30, 2026, after background calibration and before this cohort's candidate selection and deployment outcomes. It binds the source, engine revision, random-stream scheme, candidate rule, utility, scale grid, and inferential family. This is a recorded prospective freeze for the cohort, not an external preregistration or a claim that prior development played no role.

The 50-root sample size and three workload fractions are fixed features of that protocol, not values selected after a significance check. Shared roots couple the regime comparisons and keep candidate-generation variation matched. They therefore require a shared-cluster analysis. The 24-estimate family covers every cell and the planned scale, response, and interaction contrasts, rather than only the most favorable comparisons. The recorded design supplies no power guarantee for effects or populations outside that family.

The world contains six liquidity providers and twelve background takers. Each provider maintains five bid and five ask levels. Liquid-regime quotes have 100 shares per level, a two-tick half-spread, and a five-second mean refresh time; the thin regime uses 20 shares, four ticks, and 15 seconds. Background takers submit ten-share immediate-or-cancel orders on independent clocks with a 12-second mean interval and equally likely sides. A fundamental price starts at 10,000 ticks and takes symmetric one-second innovations. These settings describe a synthetic population and were not fitted to a trading venue.

A path lasts 540 seconds. The buy execution parent arrives at second 120 and makes seven decisions, 60 seconds apart, through second 480. Eighty background-only calibration paths determine expected volume during the execution window: 6,252 shares in the liquid regime and 4,239 in the thin regime. Parent size is $Q = \mathrm{round}(sV_g)$ for $s \in \{0.002, 0.05, 0.30\}$. Scale therefore denotes a fraction of calibrated background volume, not the realized participation fraction of every execution path.

Inventory response $\rho \in \{0, 1\}$ shifts a provider's quote center by a clipped function of the inventory change since parent arrival. When $\rho = 0$, this channel is disabled; executions still change book depth and later fills. Quote quantities, widths, refresh clocks, and requested submission counts are fixed across response settings. The response intervention neither represents a fully learning opponent nor disables all endogenous market effects in its zero setting.

For each of 50 roots, the frozen generator produces 16 local policy perturbations and reuses the pool in both regimes. Eight proxy backgrounds per regime evaluate every candidate against the incumbent at scale 0.05. The replay proxy executes against displayed snapshots without allowing the focal execution to change subsequent books. The recorded proxy gain and interactive utility use the same U_{100} accounting convention, although replay does not reproduce the policy-dependent book changes in deployment. The greatest average paired gain wins, with deterministic index tie breaking. A candidate is adopted only when its gain exceeds 10^{-9} bps; otherwise the procedure retains the incumbent. All 100 root–regime choices close before interactive deployment. The experiment provides no market search-pressure contrast.

Every child order has immediate-or-cancel semantics, including limited-price children; the policy class does not model a persistent passive order waiting in a queue. Candidate generation and policy equations are compatible with earlier development code, while the exchange and this cohort are separately bound. Appendix A gives further parameters and the provenance boundary.

4.2 Utility and completion preference

Let S be execution implementation shortfall in bps and f the unfinished fraction of the parent. The primary utility is

$$U_{100} = -S - 100f. \tag{12}$$

Joint interval $[L, U]$	Evidence classification	Supported interpretation
$L > 0$	Positive	Positive procedure mean under the specified utility
$U < 0$	Negative	Negative procedure mean under the specified utility
$L \leq 0 \leq U$	Unresolved	Available evidence does not determine the sign

Table 4: Evidence interpretation for a declared joint family. These classifications concern a procedure mean, not the safety of every update or an automatic deployment decision. Computation uses the recorded 10^{-9} bps tolerance around zero.

A no-fill episode has saved null shortfall, interpreted as zero in this convention, and utility -100 bps. The coefficient expresses a preference for completing the order; it is not an estimated market price of noncompletion. At fixed shortfall, one additional percentage point unfinished reduces utility by one bps. The coefficient was inherited from development and frozen before the present cohort's selections. It was not selected to make this cohort's result negative.

For a diagnostic coefficient p, keep the selected policies and their saved execution outcomes fixed. The resulting paired difference is

$$\Delta_p = (S_{\text{incumbent}} - S_{\text{selected}}) + p(f_{\text{incumbent}} - f_{\text{selected}}). \tag{13}$$

This accounting equation changes the valuation of observed execution, not the selection policy. A procedure that selected candidates using $p = 25$ could choose different policies, which these saved outcomes alone do not evaluate. Throughout the revaluation, $\mathcal{P}$ remains the procedure originally selected using $p = 100$.

4.3 Completeness and accounting checks

The cohort contains all 1,200 prescribed background checks, 100 completed choices, and 2,400 paired deployment repetitions. No failed root was replaced to reach the planned count. In 98 choices an update was adopted; the other two retain zero gain at every cell. Those two choices allow identical incumbent/selected branches to share verified episode artifacts, yielding 4,752 live episodes rather than the nominal 4,800.

Actor-specific exogenous streams exclude policy, scale, and response from their identities. Arrival-book and stream hashes are checked across paired branches. Full retained event records support a separate order-book replay and arithmetic reconciliation: conservation, price/time priority, fills, cancellations, actions, and shortfall are checked independently of the upstream matching implementation. The audit reconciles 24,837 manifest members and the original 24 estimates. These checks support faithful implementation of the synthetic design; they do not test whether its liquidity dynamics match real execution.

The engine adapter uses its own fill ledger and bounds live orders at 128. This avoids relying on an upstream cumulative-quantity statistic with a known cancellation limitation, while keeping the experiment within a tested operating range. Passing accounting checks does not validate untested simulator states or larger order populations.

5 Results: scale, utility, and uncertainty

Table 4 supplies a common reading of the joint bounds. An unresolved mean is not evidence of zero gain or safety, and a negative mean does not imply that every adopted update fails. The family must accompany the classification.

5.1 Original 24-estimate family

The four largest-scale means in Table 5 are below zero throughout their original simultaneous bands. All four largest-minus-smallest scale contrasts are also negative in that family. This supports a bounded statement about the frozen procedure at the tested large order size, rather than a claim that large updates or large trades generally fail.

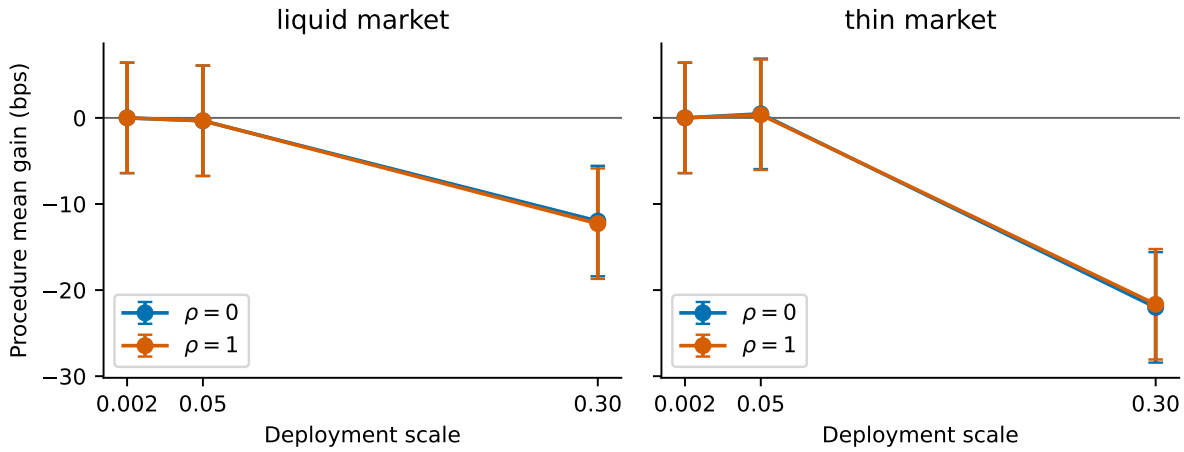


Figure 1: Procedure mean gain across the three tested order sizes under the original 100 bps utility. Error bars are approximate simultaneous 95% bands for the full 24-estimate family, resampling all 50 roots jointly across both regimes. Lines connect observations and do not imply an interpolated capacity frontier.

Regime	ρ	Mean	Simultaneous band
Liquid	0	−11.9800	[−18.3944, −5.5656]
Liquid	1	−12.2848	[−18.6992, −5.8704]
Thin	0	−22.0073	[−28.4217, −15.5929]
Thin	1	−21.6446	[−28.0590, −15.2302]

Table 5: Largest-scale procedure gains ($s = 0.30$), in bps. The original 24-estimate family uses the prespecified 100 bps completion utility and all 50 shared roots.

The two lower scales and all response and interaction contrasts remain unresolved. In particular, a negative estimate in both response conditions cannot establish that inventory response caused the loss. It also does not establish equivalent outcomes across response conditions: the contrast intervals permit differences in either direction. The complete family, including these unresolved contrasts, appears in Appendix B.

At the smallest scale the paired gains are exactly zero in the saved paths: for this policy class, an initial aggressive slice can complete the small parent. The experiment consequently certifies no positive initial scale segment. Its upper bands fail to exclude positive values at the two lower scales, but that is weaker than evidence of a profitable operating region. There is no jointly supported positive-to-negative crossing and no estimated continuous capacity threshold.

Among the 49 adopted roots per regime, estimated root means at the largest scale reverse sign in 37 liquid-regime roots and 48 thin-regime roots at each response setting. These descriptive counts summarize how widespread the observed negative root means are. They do not enter the original 24-estimate mean family or supply a population risk bound. The root-level distribution and no-update zeros are shown in Appendix C.

5.2 Proxy-to-deployment comparison at the selection scale

As a separate post-outcome diagnostic, we compare the saved proxy gain of each selected candidate with its mean paired interactive-deployment gain at $s = 0.05$, the scale used to score candidates. Both quantities use the recorded U_{100} convention, and no-update roots contribute zero to both. For root r and condition z, let $\Gamma_{r,z} = \Delta_r^{\text{proxy}} - D_{r,z,0.05}$; Table 6 reports the four root-population means. All four estimated gaps are positive and their separate four-estimate simultaneous bands exclude zero. This contrast does not identify a causal proxy bias: it combines winner selection, finite proxy-sample noise, and differences between replay and interactive books. It reuses the original 50 roots and is outside the prespecified 24-estimate family.

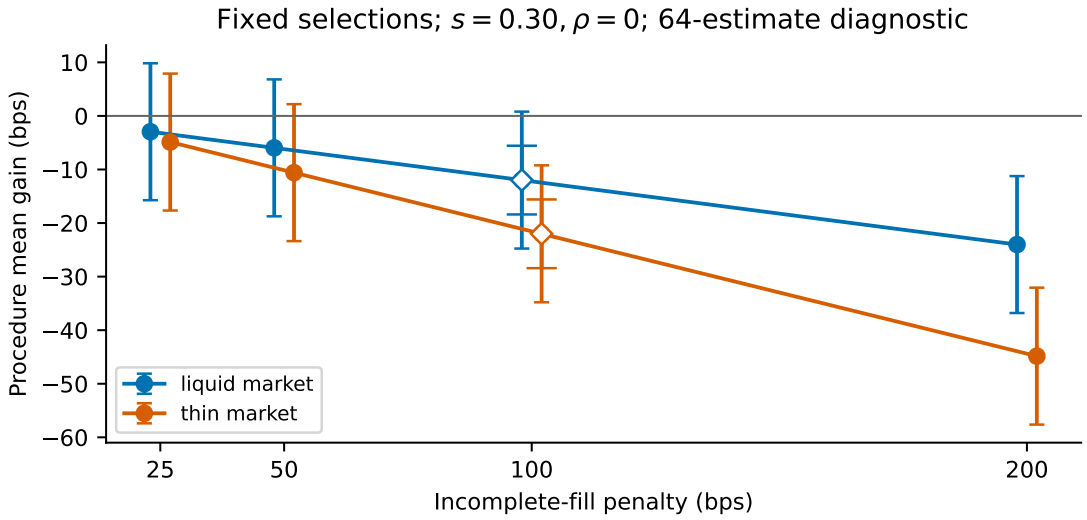


Figure 2: Largest-scale fixed-selection revaluation for $\rho = 0$. Filled markers use the 64-estimate diagnostic bands; open diamonds at 100 bps show the original 24-estimate bands. The point estimates coincide at 100 bps, while the joint uncertainty widths differ. All policies retain their original 100 bps selection.

Regime	ρ	$p = 25$	$p = 50$	$p = 100$	$p = 200$
Liquid	0	−2.95 [−15.74, 9.83]	−5.96 [−18.74, 6.82]	−11.98 [−24.76, 0.80]	−24.01 [−36.80, −11.23]
Liquid	1	−2.75 [−15.53, 10.03]	−5.93 [−18.71, 6.85]	−12.28 [−25.07, 0.50]	−25.00 [−37.78, −12.22]
Thin	0	−4.88 [−17.66, 7.90]	−10.59 [−23.37, 2.19]	−22.01 [−34.79, −9.23]	−44.84 [−57.63, −32.06]
Thin	1	−4.51 [−17.30, 8.27]	−10.22 [−23.01, 2.56]	−21.64 [−34.43, −8.86]	−44.48 [−57.27, −31.70]

Table 7: Largest-scale fixed-selection revaluation (bps), with approximate simultaneous 95% bands for the separate, outcome-aware 64-estimate family. Each entry is a mean and its band; no selection is rerun.

Regime	ρ	Proxy	Deploy.	$\widehat{\Gamma}$	Joint band
Liquid	0	0.972	-0.342	1.314	[0.649, 1.979]
Liquid	1	0.972	-0.322	1.294	[0.629, 1.959]
Thin	0	1.422	0.470	0.952	[0.288, 1.617]
Thin	1	1.422	0.365	1.057	[0.392, 1.722]

Table 6: Outcome-aware comparison at the $s = 0.05$ selection scale, in bps under the recorded U_{100} convention. Proxy, deployment, and gap means use all 50 roots; the final column gives the approximate simultaneous 95% whole-root bootstrap band for the four paired proxy-minus-deployment gaps. No-update roots contribute zero. This auxiliary family is separate from the original 24-estimate family.

5.3 Fixed-selection utility sensitivity

The revaluation uses $p \in \{25, 50, 100, 200\}$ and a separate 64-estimate family: 48 cell means and 16 largest-minus-smallest scale contrasts. It was specified after inspecting the primary outcomes. Twenty thousand shared-root bootstrap draws form its own simultaneous bands; it is neither a new independent experiment nor an expansion of the original confirmatory family.

At 100 bps, the liquid-regime point estimates are unchanged but the 64-estimate bands include zero. The thin-regime means remain negative in that analysis. At 25 and 50 bps, all four largest-scale intervals include

Regime	ρ	$\bar{a}$ (bps)	$100\bar{b}$ (pp)	$\widehat{p}^*$ (bps)	Joint set, $p \geq 0$ (bps)
Liquid	0	0.0546	-12.0346	0.454	$[0.00, 6.88]$
Liquid	1	0.4295	-12.7143	3.378	$[0.26, 18.24]$
Thin	0	0.8292	-22.8365	3.631	$[2.38, 5.36]$
Thin	1	1.1958	-22.8404	5.236	$[3.60, 7.50]$

Table 8: Largest-scale completion-utility diagnostic with original choices fixed. The threshold is a ratio of mean components, $\widehat{p}^* = -\bar{a}/\bar{b}$, computed before rounding. The sets project an approximate simultaneous 95% bootstrap rectangle for all eight component means, using 20,000 shared-root draws. This separate, outcome-aware family does not modify the original 24 or diagnostic 64 estimates.

zero; at 200 bps, all four are below zero. Thus the original result should always name its utility and inferential family. One may not choose whichever family produces the desired significance interpretation after observing its result.

5.4 Break-even completion preference: a separate diagnostic

For fixed choices, write the population mean as $\mu_{\mathcal{P}}(z; p) = a_z + pb_z$, where a_z is mean incumbent-minus-selected shortfall and b_z is mean incumbent-minus-selected unfinished fraction. If $b_z \neq 0$, its mathematical zero is $p_z^* = -a_z/b_z = -100a_z/d_z$, with $d_z = 100b_z$ in percentage points. The estimate is the ratio of the two sample means, not the mean of root-specific ratios. It concerns revaluation of the same outcomes, not reselecting candidates at each penalty or locating a capacity threshold.

Table 8 uses the unrounded root components. All four point estimates have positive shortfall gains and worse completion; their estimated zero crossings range from 0.454 to 5.236 bps. The small liquid-regime, response-off intercept is uncertain, so a positive crossing is not established in that cell. Shortfall improvement alone consequently does not determine the value of a procedure that changes completion.

Ratio uncertainty requires care. We resample all 50 roots jointly and construct a fixed-standard-error maximum-deviation rectangle for the eight component means, then project it onto $p \geq 0$ where $a + pb = 0$. The four resulting sets are bounded here, but the construction can return an unbounded or empty set. Each liquid cell has three denominator-sign changes among 20,000 draws; those draws remain in the component bootstrap and no percentile ratio interval is reported for those cells. Appendix D gives the construction and its limits. This outcome-aware eight-component family is separate from both earlier analyses; its intervals cannot replace their unresolved findings or be treated as new confirmation.

5.5 Root sensitivity without new confirmation

Post-outcome diagnostics remove each of the 50 roots in turn and reconstruct the original 24-estimate bands with 49 roots. Four additional constructions use two alternative random seeds, a standardized whole-root maximum, and a standardized Gaussian root-multiplier maximum. Their details and full outputs are supplied in Appendix C.

All four largest-scale upper bounds stay negative across these checks; the closest alternative upper bound to zero is -3.13 bps. This reduces concern that a single observed root or the tested resampling choices alone account for the original finding. It does not address misspecified liquidity, utility preferences, or other candidate classes. Because the diagnostics were chosen after the outcome and share the same data, no joint coverage is asserted across all methods or deletions, and no additional independent confirmation is claimed.

6 Practical implications: a deployment assessment report

The assessment specification can be used as a compact report schema for another frozen update procedure. It is a reporting and audit structure, not an automatic deployment gate. A report should bind the fields in Table 9 before interpreting outcome data and should assign a separate identity to any outcome-aware diagnostic.

Field	Required declaration	Market instantiation
Procedure	Candidate generation, selection rule, incumbent	Frozen generator; 16 candidates; replay-proxy winner
Adoption	Replacement gate and no-update behavior	Adopt only above 10^{-9} bps; retain incumbent otherwise
Population	Independent unit and target population	50 shared generation roots; 100 root–regime decisions
Deployment	Environment, scale, and response conditions	Synthetic matching engine; two regimes, three scales, two responses
Utility	Outcome terms and preference parameters	$-S - pf$; primary $p = 100$ bps
Costs	Research and operational costs included?	Not measured: compute, API/token use, latency, safety, or long-horizon failure costs
Evidence	Inferential family and sensitivity identity	24-estimate bands; separate 64-estimate, proxy-gap, and break-even diagnostics
Interpretation	Positive, negative, unresolved, and exclusions	Sign classification for the declared procedure mean

Table 9: Procedure-level deployment assessment report schema. The final column records this study's choices; another application must supply its own procedure, population, utility, and evidence family.

The evidence labels in Table 4 are useful only relative to the declared utility and inferential family. Positive evidence for a procedure mean does not establish acceptable tail risk or safe deployment, and an unresolved cell is an evidence gap rather than evidence of zero value. The bounded planner-code case is retained in the Supplementary Material as an auxiliary example with three generation roots and a fixed task panel; it is not pooled with the market cohort or used to claim agent generalization.

For future applications, a net utility could subtract research and operations costs, for example $V_{\text{net}} = \mu_{\mathcal{P}} - \lambda_c C_{\text{compute}} - \lambda_t C_{\text{latency}} - \lambda_r C_{\text{risk}}$. The present study does not observe these terms, so it reports deployment utility conditional on the declared execution preference rather than a net economic value.

7 Discussion and reproducibility

7.1 Implications for machine-learning deployment evaluation

The results illustrate why deployment utility is a property of a specified update-selection procedure rather than an intrinsic property of a proxy score or selected policy. The same frozen procedure can receive different assessments when operating conditions or completion preferences change. A deployment claim is also inseparable from its inferential target: repeated evaluations within a root do not replace independent update decisions, and a wider inferential family can leave a previously resolved conclusion uncertain.

This perspective is relevant to policy-learning and iterative optimization systems beyond the execution setting studied here. Extending it to coding agents, autonomous research systems, or other operational environments requires independently frozen procedures, domain-appropriate utilities, explicit cost and failure terms, and new evaluation populations. The current market evidence does not establish that the same effects occur in general-purpose agents.

What the release reproduces. The anonymous package contains normalized saved observations, pinned numeric references, source/protocol identifiers, and manuscript sources. Its single wrapper recomputes the 24- and 64-estimate analyses, break-even diagnostic, root audits, and supplementary planner summaries before generating tables and vector figures and compiling the paper. Numeric differences are checked within 10^{-9} in native units. The package is tested by extraction into a separate directory and execution in an isolated environment; rebuilding PDF bytes is not the criterion because rendering metadata can differ.

This is saved-outcome reanalysis. It does not regenerate model proposals, re-fit the classification tasks from original datasets, or reconstruct all exchange episodes. Those earlier raw-record and grade audits have

separate receipts. Successful reanalysis establishes arithmetic and packaging reproducibility, while an external rerun of the original scientific processes remains a stronger test. Failure evidence and incomplete studies in the wider project are preserved but are not evidence for the current claims.

8 Limitations and operational scope

Statistical limits. The independent-market root population is finite in this sample and synthetic in its definition. The bootstrap supplies approximate conditional joint uncertainty, not exact coverage or simulator-model uncertainty. The adopted-root sign counts additionally reflect finite deployment replication. Utility revaluation and root diagnostics are outcome-aware. No selection-policy comparison at alternative penalties, market search-pressure effect, monotone scaling law, or equivalence of response conditions is established.

External validity. The matching engine's priority rules do not make the surrounding market realistic. Provider refresh, depth, and inventory response follow a designed model; persistent execution queues, institutional constraints, and real own-order impact are unvalidated. The experiment does not justify professional trading or calibration to a particular crypto venue. The planner case covers bounded edits and a fixed task panel, with no independent model-backbone replication or completed primary autonomous-agent study.

The assessment does not measure API costs, token consumption, computational latency, agent safety, or long-horizon reliability. Completion utility is an explicit preference over execution outcomes, not a surrogate measurement of these other dimensions. Referencing broader deployment-evaluation work does not make them part of the present evidence.

Practical use. A deployment report should name the procedure being assessed, retain no-update decisions, declare utility before interpreting value, and present uncertainty over the actual intended claim. Positive evidence on a tested grid still leaves operational failure severity and deployment conditions to be assessed. Conversely, unresolved cells are evidence gaps, not proof of zero benefit. Any future extension to a real workflow requires a separately frozen evaluation and appropriate cost and failure accounting; this paper provides no ready-made authorization for deployment.

9 Conclusion

This study evaluates the deployment utility of a frozen policy-update procedure rather than the isolated proxy performance of a selected candidate. Its central finding is conditional: under one synthetic execution environment, prespecified completion preference, and original joint inferential family, the procedure has negative mean utility at the largest tested order size. Lower-scale cells and response contrasts remain unresolved. Fixed-selection revaluation demonstrates that utility preferences and inferential scope must accompany any deployment statement, while the break-even analysis provides an explicitly post-outcome characterization of the same saved outcomes. A separate selection-scale diagnostic finds positive proxy-minus-deployment gaps across all four regime–response cells, but its outcome-aware construction cannot identify a causal proxy-bias effect.

The methodological implication is to distinguish the value of an individual update from the expected utility of the complete procedure that generates and adopts updates. A defensible assessment retains incumbent decisions, declares operating conditions and utility, identifies independent evaluation units, and aligns statistical evidence with the intended claim. The formal properties supplied here are reusable characterizations, not a universal deployment theorem or a new fidelity algorithm.

The evidence is conditional on one frozen synthetic market and one update procedure. It does not establish general agent reliability, professional-market execution validity, a universal scaling law, or net value after compute and operational costs. Independent simulators, alternative update procedures, and autonomous-agent populations remain necessary for broader validation.

Author Contributions and AI Assistance. The authors conducted the research and prepared the manuscript. AI tools assisted with manuscript drafting and code development. The authors reviewed all AI-assisted outputs and take full responsibility for the final work.

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A Protocol details and provenance

The engine revision is `f94150294a85d7b415ca4518590b5a661d6f9958`. It uses integer prices, quantities, and microsecond event times. Fundamental innovations are $-1, 0, +1$ ticks with probabilities $0.25, 0.50, 0.25$. Provider quote centers add

$$\mathrm{clip}(-\,\mathrm{round}\,\{2\rho\,[I_j(t) - I_j(120)]/100\}\,, -8, 8)$$

ticks to the current fundamental price; rounding is ties-to-even. All actor cash and inventory begin at zero, with shorting and negative cash explicitly permitted. This convention is part of the simulation, not an account-risk model.

The event phases order fundamental changes, provider refreshes, background takers, activation, proxy snapshots, and execution children. Provider refresh cancels residual orders and inserts the fixed ladder. Limited-price execution children cancel any unfilled residual within the atomic operation. Separate cash/inventory ledgers attribute both sides of each fill. The retained events support checking cancellations against actual remaining quantity and reconstructing book state without invoking the upstream matcher.

Calibration and engineering roots are separate from the 50 prospective roots, locally labeled 8101–8150 within cohort `cohort-20260930-v1`. Regime-specific expected volumes come from 40 calibration paths per regime. The protocol fixes eight proxy paths and four deployment paths per root/regime. Paired deployment paths share exogenous innovations and arrival states, while policies and resulting endogenous event paths may differ.

Exact-zero and small-scale checks test the implementation's boundary behavior and are not substitutes for new independent roots.

The labels 8101–8130 also appear as seeds in the earlier PIVOT E2C cohort and are not globally unique root identifiers. PIVOT E2C uses NumPy `default_rng` for candidate and environment draws. Here candidate perturbations use Python `random.Random`; deployment streams use SHA-256-derived seeds namespaced by cohort, root, split, replicate, actor, and channel before using `random.Random`. The submitted PIVOT data and figure files have no byte-identical matches to the current outcome files and figures. The shared integer labels therefore do not identify shared saved outcomes; the distinct cohort protocols, rather than label novelty, define the current sample.

The 100 bps completion preference, policy equations, and candidate perturbations were carried forward from development to maintain a fixed comparison. The new exchange lineage, prospective roots, and source freeze distinguish the principal cohort from those development trials. They do not establish independence from every design decision informed by development. The prior PIVOT supplement's virtual finance fixtures use a different execution construction and are not reported as current results.

The supplemental provenance inventory identifies the original protocol and the saved inputs used by the portable release. The actual ICLR main manuscript and its submitted ZIP are the related-work comparison baseline, alongside the public preprint. Their identities are retained internally; the anonymous main manuscript is included only as comparison material. Reuse of a statistical tool with attribution is distinguished from reporting an existing empirical result as a new contribution.

B Complete original inferential family

Table 10 preserves the full prospective family. The four scale contrasts subtract the smallest-scale cell from the largest-scale cell within each regime and response setting. Response contrasts subtract $\rho = 0$ from $\rho = 1$ at a fixed scale. An interaction subtracts the $\rho = 0$ scale contrast from the $\rho = 1$ scale contrast. All share the same root resampling and maximum-deviation radius.

C Post-outcome root diagnostics

C.1 Construction and deletion results

These checks reuse the frozen selections and 100 bps utility after the original outcomes were inspected. They neither add independent roots nor extend the original confirmatory family. Each method resamples the complete root vector across both regimes and all 24 estimands.

Standardization divides bootstrap deviations by the original sample standard error, not a reestimated standard error in each draw. Gaussian multipliers approximate the centered root-mean distribution using its empirical covariance. Constant observed columns are flagged; a zero empirical standard error does not establish zero population uncertainty. Numerical bands remain approximate and conditional on the frozen synthetic world.

The standardized maximum uses the original sample standard error for each estimand, not a standard error recomputed within every bootstrap draw. The Gaussian multiplier draws centered root vectors with conditional mean-distribution covariance matched to the sample covariance divided by the root count. These alternatives probe particular construction choices. They are not independent replications, and reporting the greatest upper bound among the checks does not create a new simultaneous confidence procedure.

D Break-even diagnostic construction

The November 4 diagnostic specification records that the v5 outcomes and component means had already been inspected. At $s = 0.30$, each root contributes $(a_{r,j}, b_{r,j})$ for each of the four regime–response cells j, averaging its four paired repetitions and retaining no-update zeros. Let $X_r \in \mathbb{R}^8$ collect these components, and let $\widehat{\sigma}_k$ be each sample mean's standard error. Twenty thousand common root resamples, using PCG64

Regime	Estimand	Mean	Simultaneous band
Liquid	Cell: $\rho = 0, s = 0.002$	0.0000	[−6.4144, 6.4144]
Liquid	Cell: $\rho = 0, s = 0.05$	−0.3420	[−6.7564, 6.0724]
Liquid	Cell: $\rho = 0, s = 0.3$	−11.9800	[−18.3944, −5.5656]
Liquid	Cell: $\rho = 1, s = 0.002$	0.0000	[−6.4144, 6.4144]
Liquid	Cell: $\rho = 1, s = 0.05$	−0.3219	[−6.7363, 6.0925]
Liquid	Cell: $\rho = 1, s = 0.3$	−12.2848	[−18.6992, −5.8704]
Thin	Cell: $\rho = 0, s = 0.002$	0.0000	[−6.4144, 6.4144]
Thin	Cell: $\rho = 0, s = 0.05$	0.4695	[−5.9449, 6.8839]
Thin	Cell: $\rho = 0, s = 0.3$	−22.0073	[−28.4217, −15.5929]
Thin	Cell: $\rho = 1, s = 0.002$	0.0000	[−6.4144, 6.4144]
Thin	Cell: $\rho = 1, s = 0.05$	0.3650	[−6.0494, 6.7794]
Thin	Cell: $\rho = 1, s = 0.3$	−21.6446	[−28.0590, −15.2302]
Liquid	Scale: $\rho = 0$	−11.9800	[−18.3944, −5.5656]
Liquid	Scale: $\rho = 1$	−12.2848	[−18.6992, −5.8704]
Thin	Scale: $\rho = 0$	−22.0073	[−28.4217, −15.5929]
Thin	Scale: $\rho = 1$	−21.6446	[−28.0590, −15.2302]
Liquid	Response: $s = 0.002$	0.0000	[−6.4144, 6.4144]
Liquid	Response: $s = 0.05$	0.0201	[−6.3943, 6.4345]
Liquid	Response: $s = 0.3$	−0.3048	[−6.7192, 6.1096]
Thin	Response: $s = 0.002$	0.0000	[−6.4144, 6.4144]
Thin	Response: $s = 0.05$	−0.1046	[−6.5190, 6.3098]
Thin	Response: $s = 0.3$	0.3627	[−6.0517, 6.7771]
Liquid	Scale-by-response	−0.3048	[−6.7192, 6.1096]
Thin	Scale-by-response	0.3627	[−6.0517, 6.7771]

Table 10: All 24 prospective estimates in bps, using one shared-root maximum-deviation band. Scale denotes $0.30 - 0.002$; response denotes $1 - 0$. Unresolved contrasts are retained.

Regime	ρ	LOO mean range	Max. LOO upper	Max. alternative upper
Liquid	0	[-13.26, -11.49]	-4.98	-3.13
Liquid	1	[-13.50, -11.78]	-5.30	-3.62
Thin	0	[-23.01, -21.83]	-15.28	-15.61
Thin	1	[-22.63, -21.47]	-14.92	-15.25

Table 11: Largest scale $s = 0.30$: all values are bps. Leave-one-root-out (LOO) checks refit a 24-estimate unstandardized maximum-deviation band using 49 shared roots and 20,000 draws for each of 50 deletions. Alternative bounds use two independently seeded paired-root runs, a fixed-sample-SE standardized paired-root maximum, and a standardized Gaussian root-multiplier maximum, each with 20,000 draws. No joint coverage is claimed across methods or deletions.

seed 20261104, give the 95th percentile of

$$\max_{k \le 8} \frac{|\overline{X}_k^{(b)} - \overline{X}_k|}{\widehat{\sigma}_k}.$$

The critical value is 2.512769. The denominators are the original sample standard errors, not bootstrap-specific estimates. All eight empirical variances are positive; the implementation rejects constant components rather than treating zero empirical variation as known population certainty.

For a cell's rectangle $[a_L, a_U] \times [b_L, b_U]$, the projected set is

$$\mathcal{C}_j = \{p \ge 0 : a_L + p b_L \le 0 \le a_U + p b_U\}.$$

This inversion uses the affine utility and preserves unbounded or empty sets when the data permit them. Conditional on coverage of the eight-component rectangle, every existing nonnegative zero of those covered

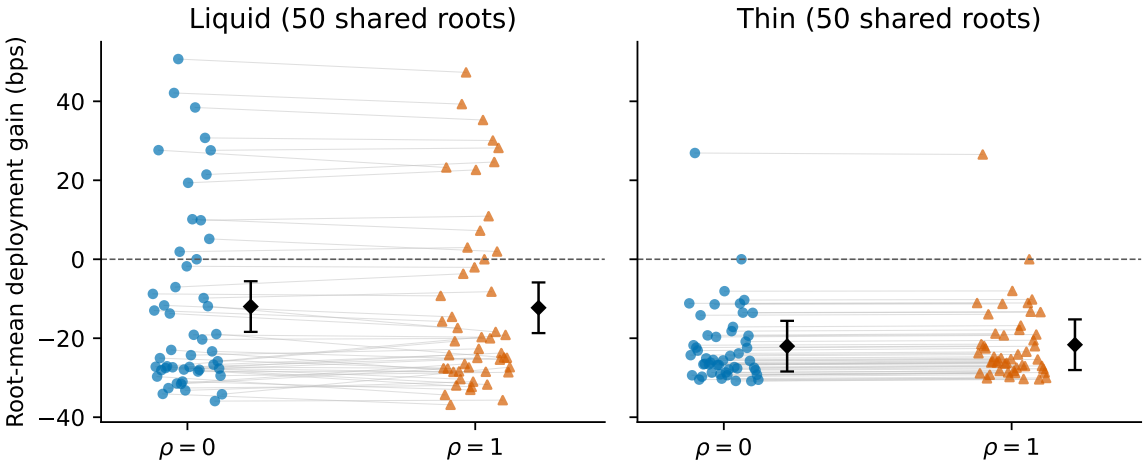


Figure 3: Fifty root-level procedure gains at the largest scale under the original 100 bps utility. Each point averages four paired deployments; lines connect the same root across response settings. Incumbent-retaining roots contribute zero. Diamonds and bars show the original full-cohort means and 24-estimate simultaneous bands, not uncertainty for individual roots.

mean utilities belongs to its projected set. Coverage is approximate and conditional on the synthetic population; a nonempty projected set does not prove that a nonnegative zero exists. The nominal level treats this diagnostic specification as given and does not adjust for choosing it after inspecting outcomes. The rectangle discards some within-cell covariance geometry during projection, so the sets can be conservative. No joint guarantee is asserted across this analysis and the earlier 24- or 64-estimate families.

For both liquid cells, three resampled denominators change sign; none is zero. For both thin cells, neither event occurs. The report retains these counts, all draws in the component construction, and the untrimmed pointwise percentile ratios for the two stable thin cells (3.113–4.151 and 4.702–5.768 bps). Those narrower pointwise summaries are not the joint sets in Table 8. The liquid, response-off intercept rectangle includes zero and negative values; its displayed $[0, 6.88]$ set therefore does not establish a positive-to-negative population crossing. These choices follow the separate outcome-aware diagnostic specification and do not change the frozen original analysis.

E Adoption and utility interpretation

Equations 1–3 separate a procedure's expected gain from the outcomes of its adopted subset. Conditioning on observed deployment outcomes to decide adoption would change the estimand and invalidate the stated gate. Here selection is closed before those outcomes. Changing the utility coefficient while holding selection fixed is also distinct from reevaluating a procedure that would optimize the new coefficient.

In particular, a positive conditional gain given adoption can coexist with a small procedure mean when adoption is rare; a given reversal rate can coexist with either positive or negative value depending on the loss magnitudes. Neither fact is offered as a new theorem. These accounting distinctions specify what a saved-outcome analysis can answer. Missing, failed, or altered root records must not be silently converted to no-update zeros; the portable checks reject incomplete identities and nonfinite outcomes.

Supplementary material

Reproduction, diagnostics, and scope

Complete research package
5 November 2026
Release 202611050855

Supplementary Material for Evaluating the Deployment Utility of a Frozen Policy-Update Procedure: Scale, Completion Preference, and Joint Evidence

Anonymous authors
Paper under double-blind review

1 Evidence and independent units

The principal cohort contains 50 independent roots, each shared across two synthetic liquidity regimes. The 100 selection decisions and 2,400 paired deployment repetitions are repeated observations within those roots. The original family contains 24 estimates under a 100 bps incomplete-fill penalty. Root resampling carries both regimes and every cell together. All results remain conditional on the frozen policy generator, selection rule, exogenous population, and matching-world implementation.

The utility diagnostic retains the original policies and outcomes and varies only their valuation at 25, 50, 100, and 200 bps. Its 64-estimate family was defined after the primary outcomes were inspected. Root deletion and alternative-band constructions also reuse the original data after inspection. None of these checks supplies another independent cohort or changes the original 24-estimate family.

The proxy-to-deployment comparison is a separate post-outcome diagnostic at the $s = 0.05$ selection scale. It contrasts the selected candidate's saved proxy gain, with no-update decisions set to zero, against the mean of four paired interactive deployment gains under the recorded U_{100} convention. Its four root-level differences use a separate whole-root simultaneous band. This comparison does not identify causal proxy bias and is not part of the original 24-estimate family.

The additional November 4 break-even diagnostic uses the two saved utility components at the largest scale. Its separate specification acknowledges inspection of the earlier results. The eight-component whole-root bootstrap rectangle and its projected nonnegative zero sets retain every root, including no-update zeros. The analysis fixes all original choices; it does not assess re-selection at other completion penalties. Its uncertainty family is separate from the 24- and 64-estimate families.

The planner study contains three generation roots with 32 ordered attempts each and a fixed 20-task deployment panel. Invalid/duplicate attempts retain their positions. Its positive observed means are an auxiliary case with exploratory pointwise uncertainty; they are not evidence for unrestricted agents or an independent model backbone. The full case is reported here rather than in the main evidence section, and its outcomes are not pooled with market roots.

2 What is inherited and what is separately evidenced

Improvement Fidelity, reversal summaries, no-update decisions, paired comparisons, and root-bootstrap tools are attributed to related work (Anonymous, 2026) or used as standard statistical operations. PIVOT-KG, its validation-allocation comparisons, the prior response experiments, and its virtual finance fixtures are not contributions of this submission. The primary new evidence is the separately frozen matching-engine cohort; the planner-code study is retained as a bounded supplementary case, with explicitly post-outcome diagnostics.

The matching engine has a different source lineage from the earlier development exchange. Policies, candidate equations, and the completion utility were nevertheless carried forward for compatibility. Implementation

lineage does not imply independence from all development choices. The engine's accounting and independent replay checks validate saved synthetic events rather than real-market realism.

An anonymous comparison copy of the actual concurrently reviewed conference manuscript is supplied as `related_work/pivot-anonymous.pdf`. It is solely related-work material, not part of the new result set. The originality review also used that submission's separate supplementary archive, which includes older source and virtual execution examples. Its older source version was not silently substituted for the actual conference PDF. The full public reference is retained outside the anonymous review package for restoration after review.

3 Portable reanalysis and build

The ZIP contains manuscript sources, normalized outcome data, expected summaries, a data dictionary, dependencies, and a hash inventory. Install `reproduction/requirements.txt` into an isolated Python environment and run, from the package root:

```
bash reproduce_all.sh /path/to/output
```

The system additionally needs a working TeX installation with `latexmk`, `pdflatex`, and `bibtex`. The Python dependencies are pinned in the package; no model credentials, simulator, original dataset download, or task-fitting runtime is needed for this saved-outcome check.

The wrapper verifies the outcome hashes; reconstructs the original 24 estimates, the 64 diagnostic estimates, the four proxy-to-deployment gaps, and the five planner-prefix summaries; recomputes all 50 root deletions, four alternative bands, and the break-even diagnostic; then generates the tables and vector plots consumed by the manuscript builds. All numeric results are checked against the bound references within 10^{-9} in their native units. The build uses the recomputed outputs, not a precompiled PDF. PDF text and numerical agreement are checked separately from rendering bytes.

The break-even output includes the 50-root component data, the eight simultaneous component intervals, the nonnegative zero sets, and denominator-instability counts. No ratio draws are silently dropped. In the two liquid cells, denominator sign changes suppress the simple percentile-ratio summary; the projected component sets remain available. The diagnostic script handles bounded, unbounded, and empty sets and rejects constant empirical components for this standardized construction. Its methods and limitations are detailed in the main-paper appendix.

The original market CSV includes the independent root, regime, adopted candidate, incumbent, proxy gain, paired deployment gain, scale, response, replication index, no-update flag, configuration hash, and the two saved utility components. A complete record key is the tuple of root, regime, response, scale, and repetition. Missing/duplicate keys, nonfinite values, inconsistent selections, nonzero no-update gains, and failed statuses are rejected. A failed attempt is never repaired by replacing the root.

Portable reanalysis does not regenerate model responses, original task fits, or exchange event streams. The raw-event replay and original grade audits are distinct retained evidence. The package establishes that its saved-outcome claims can be reconstructed; it is not a new independent scientific replication. Uncompleted wider-project studies remain excluded from the paper and preserved in the internal archive.

4 Planner visualization and scope

Nested prefixes reuse the same generation sequence, including invalid attempts. Selection uses four proxy tasks and a no-update option. A fixed menu and at most two training/finish actions bound the candidate programs. These constraints make the case interpretable but sharply limit claims about tool-using or autonomous research agents. No shared numerical stress scale is assumed between this study and market execution.

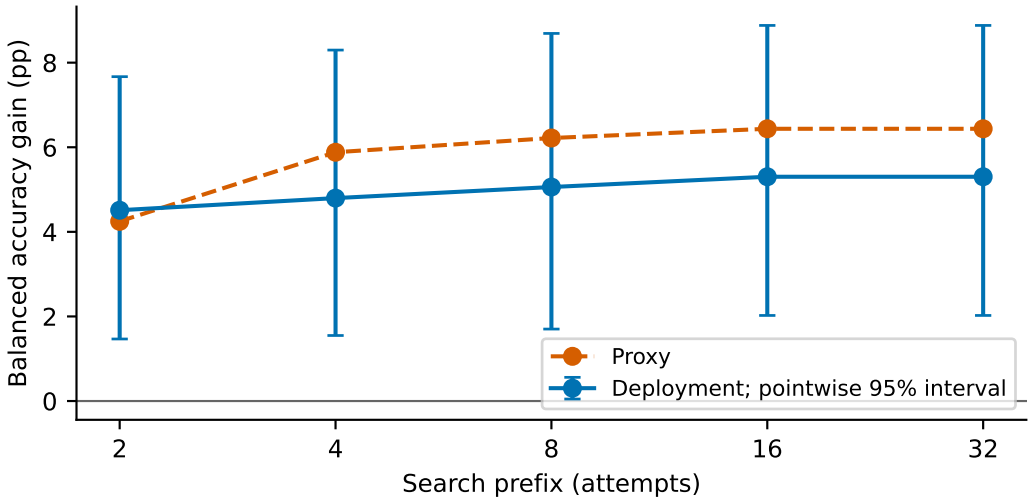


Figure 1: The bounded planner-code case. Gains are balanced-accuracy percentage points. Deployment intervals are exploratory crossed root/task bootstrap intervals over three generation roots and the fixed 20-task panel. The intervals are pointwise and do not test the difference between prefixes.

5 Review and provenance limits

The original freeze, implementation audit, saved-outcome reanalysis, and current manuscript audit are distinct records. A new scope/version binds the present manuscript without rewriting historical experiment receipts. The absence of duplicate file hashes or long matching prose passages alone is not proof of semantic originality; the comparison also considers the claims, cohorts, methods, and result sources.

Final authorship, conflicts, rights, and submission declarations require author confirmation. The paper's bounded scientific claims do not certify completion of the broader project or guarantee a venue decision. The review materials maintain anonymous authors and cite related work substantively.

References

Anonymous. When Better Gets Worse: Improvement Fidelity for Self-Improving Agents in Adaptive Worlds, 2026. Related manuscript under review; anonymized comparison copy supplied in the supplementary source package.

Classic Beamer slides

Research presentation

Complete research package
5 November 2026
Release 202611050855

Evaluating the Deployment Utility of a Frozen Policy-Update Procedure: Scale, Completion Preference, and Joint Evidence

Research Presentation

Scope of This Study

Empirical and methodological evaluation of one frozen replay-proxy policy-update procedure in a controlled synthetic continuous double auction, using saved outcomes, root-level inference, and explicit distinctions between confirmatory and post-outcome diagnostics.

Why Proxy Scores Are Insufficient for Deployment Assessment

- Candidate policy updates are frequently screened on cheap replay or proxy benchmarks before interactive deployment.
- A positive proxy gain does not guarantee positive incumbent-relative utility once the selected policy interacts with an active environment.
- Four distinct mechanisms can separate proxy screening from deployed utility:
  1. **Winner selection:** choosing among multiple candidates induces selection optimism on finite proxy samples.
  2. **Interactive feedback:** deployed orders alter book state, liquidity, and subsequent fills unlike passive replay.
  3. **Operating scale:** order size changes market impact and execution completion rates.
  4. **Completion preferences:** deployment utility depends on how incomplete fills are penalized relative to execution prices.
- Sign-reversal frequency alone is insufficient: deployment value depends on the magnitudes of gains and losses, including no-update decisions.

Procedure-Level Estimand and Research Questions

Formal Target (Procedure-Level Deployment Utility)

Expected incumbent-relative utility of the *complete frozen procedure*—retaining both candidate adoption and no-update decisions—under declared operating conditions, candidate-generation population, utility function, and joint evidence family.

What this paper does not propose:

- No new off-policy estimator, fidelity metric, selection algorithm, or universal deployment theorem.

Primary Research Questions:

1. **Operating scale & liquidity:** How does deployed incumbent-relative utility vary across order scales ($s \in \{0.002, 0.05, 0.3\}$), liquidity regimes (liquid, thin), and inventory responses ($\rho \in \{0, 1\}$)?
2. **Utility and evidence sensitivity:** How does fixed-selection revaluation across incomplete-fill penalties ($p \in \{25, 50, 100, 200\}$ bps), together with the choice of inferential family, change the deployment assessment?

Auxiliary diagnostic: a separate post-outcome proxy-to-deployment comparison is reported at the selection scale ($s = 0.05$); it is not a primary research question.

Related Work and Scope Boundary

Research Area	Relationship and Boundary to This Work
Off-Policy Evaluation (OPE) & High-Confidence OPE	Evaluates policies from logged data. We evaluate paired interactive deployments of a frozen replay-screened update procedure.
Safe Policy Improvement (SPIBB)	Constructs safer policy improvements under uncertainty. We audit a fixed procedure without altering its selection rule.
Conservative Evaluation of Offline Policy Learning	Close TMLR precedent on rigorous evaluation of offline learning pipelines. We focus on procedure-level deployment utility across scale and penalty preferences.
Active Offline Policy Selection	Adaptively allocates limited online evaluation budgets across candidates. Here, selection is frozen on replay proxies prior to deployment assessment.
HAL & Agent Reliability	Studies system-level benchmarking and ranking measurement. We evaluate a specific frozen market-execution update procedure.
PIVOT (Improvement Fidelity & PIVOT-KG)	Studies transition-level Improvement Fidelity and PIVOT-KG. This paper evaluates frozen procedure-level deployment utility and does not replace those methods or repeat PIVOT results.

Formal Characterization 1: Reversal Accounting

These are elementary formal characterizations, not new estimators or deep general theorems.

Gain and loss magnitudes matter

Let A denote adoption; an incumbent-retaining decision has zero procedure gain. Then

$$\mathbb{E}[\Delta U] = \Pr(A)[\Pr(\Delta U \geq 0 \mid A)\mathbb{E}[\Delta U \mid A, \Delta U \geq 0] - \Pr(\Delta U < 0 \mid A)\mathbb{E}[|\Delta U| \mid A, \Delta U < 0]] \,.$$

Interpretation

Adoption or reversal frequency alone cannot determine procedure value. Two procedures can have the same reversal rate but opposite means when their positive and negative outcome magnitudes differ.

The estimand retains no-update decisions; it is not the mean of adopted updates only.

Formal Characterizations 2–3: Utility and Sampling

2. Fixed-selection completion-utility frontier

For the same selected policies and deployments, separating the execution component a from completion shortfall b gives

$$\mu(p) = a + p\,b.$$

A mathematical zero, when $b \neq 0$, is $p^* = -a/b$. Changing p does not reselect candidates.

3. Root/deployment variance decomposition

For R independent roots and m repeated deployments per root,

$$\mathrm{Var}(\widehat{\mu}) = \frac{\mathrm{Var}_R(g_R)}{R} + \frac{\mathbb{E}_R[\sigma_R^2]}{Rm}.$$

More repetitions reduce within-root deployment noise; only more independent roots reduce root-population variation.

Why it matters here

The 2,400 paired deployment records improve within-root precision, but the primary independent unit remains the 50-root cluster.

Frozen Experimental Protocol and Statistical Units

Frozen selection and evaluation design

- **Environment:** Synthetic continuous double auction.
- **Procedure:** One frozen replay-proxy policy-update procedure.
- **Generation roots:** 50 independent generation roots; each root screens 16 candidate policies.
- **Selection decisions:** 100 root–regime selection decisions: one for each of 50 roots in each liquidity regime.
- **Interactive evaluation:** 2,400 paired interactive deployment records; selected policies and incumbents are evaluated on separate interactive books.

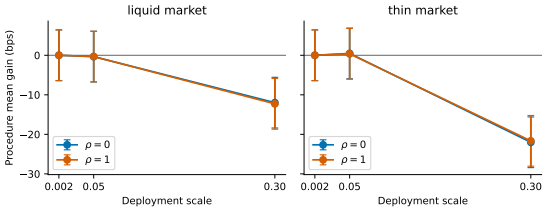
Operating grid and confirmatory family

- **Liquidity regimes:** Liquid and thin.
- **Order scales:** $s \in \{0.002,\ 0.05,\ 0.3\}$.
- **Inventory response:** $\rho \in \{0,\ 1\}$.
- **Confirmatory utility:** U_{100} with a 100-bps incomplete-fill penalty.
- **Multiplicity control:** Approximate simultaneous confidence band over a prespecified 24-estimate confirmatory family.

Independent unit of inference

The independent statistical unit is the **root** ($R = 50$), not each deployment record. Repeated deployments reduce within-root noise without adding new roots.

Main Scale Results Under U_{100}



Largest-Scale ($s = 0.3$) Cell Means (bps)

Original 24-estimate simultaneous intervals:

Cell ($s = 0.3$)	Mean	Simultaneous CI
Liquid, $\rho = 0$	-11.9800	$[-18.3944, -5.5656]$
Liquid, $\rho = 1$	-12.2848	$[-18.6992, -5.8704]$
Thin, $\rho = 0$	-22.0073	$[-28.4217, -15.5929]$
Thin, $\rho = 1$	-21.6446	$[-28.0590, -15.2302]$

- At $s = 0.3$, all four regime-by-response cells exhibit clearly negative incumbent-relative deployment utility.
- Thin liquidity doubles the magnitude of the largest-scale loss relative to liquid books.

What the Prespecified 24-Estimate Joint Family Supports

Supported by Original Joint Family

- Negative incumbent-relative deployment utility under U_{100} at the largest order scale ($s = 0.3$) in all four operating cells:
  - Liquid, $\rho = 0$: -11.9800 bps
  - Liquid, $\rho = 1$: -12.2848 bps
  - Thin, $\rho = 0$: -22.0073 bps
  - Thin, $\rho = 1$: -21.6446 bps
- Every corresponding simultaneous confidence interval excludes zero by at least 5.56 bps.

Unresolved Under Original Joint Family

- Lower-scale cell means at $s = 0.002$ and $s = 0.05$ are not resolved as strictly positive or negative under the 24-estimate simultaneous band.
- Inventory-response contrasts ($\rho = 1$ vs. $\rho = 0$) and interaction contrasts remain unresolved under the original family.
- Absence of a resolved effect in these smaller-scale or contrast cells is treated as inconclusive, not as proof of zero effect.

Post-Outcome Proxy-to-Deployment Gap Diagnostic ($s = 0.05$)

Outcome-Aware Diagnostic Specification

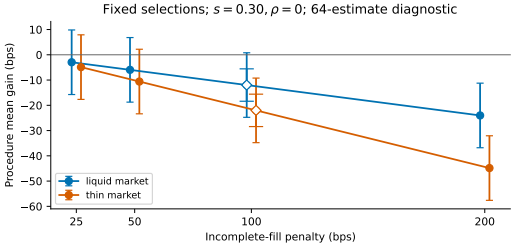
Compares saved replay-proxy gains (with no-update decisions set to zero) against paired interactive deployment gains at the selection scale $s = 0.05$, using a separate whole-root simultaneous interval construction.

Cell	Gap (bps)	Whole-root simultaneous interval
Liquid, $\rho = 0$	1.314	[0.649, 1.979]
Liquid, $\rho = 1$	1.294	[0.629, 1.959]
Thin, $\rho = 0$	0.952	[0.288, 1.617]
Thin, $\rho = 1$	1.057	[0.392, 1.722]

Explicit Methodological Caveats:

- **Post-outcome:** Separate diagnostic, outside the original 24-estimate confirmatory family.
- **Composite gap:** Combines winner selection across 16 candidates, proxy-sample noise, and replay–interactive differences.
- **No causal decomposition:** Not an isolated or causal estimate of proxy bias.

Post-Outcome Completion Preference Sensitivity and Break-Even



Post-Outcome Diagnostic Scope

- Revalues the *same frozen selected policies* at incomplete-fill penalties $p \in \{25, 50, 100, 200\}$ bps.
- Does **not** reselect candidates under alternative penalties.
- The 64-estimate expanded family, root-deletion audits, alternative simultaneous bands, and break-even analysis are all **post-outcome diagnostics** separate from the confirmatory 24-estimate family.

Affine break-even diagnostic

Uses $\mu(p) = a + p\,b$ with a separate joint component construction. It is descriptive and does **not** establish a confirmatory threshold.

Specification for an Auditable Deployment Assessment Report

To make procedure-level evaluation reproducible and transparent, an assessment report should explicitly separate frozen design choices from post-outcome sensitivity checks:

Report Component	**Required Specification in This Protocol**
1. Procedure & Population	Frozen update rule with no-update retention; 50 roots, 16 candidates per root, and 100 root–regime decisions.
2. Operating Grid	Explicit environment regimes (liquid, thin), order scales ($s \in \{0.002, 0.05, 0.3\}$), and inventory responses ($\rho \in \{0, 1\}$).
3. Utility Accounting	Primary U_{100} with a 100-bps incomplete-fill penalty; 2,400 paired interactive records.
4. Unit of Inference	Root-level clustering ($R = 50$) recognizing that repeated deployments reduce within-root variance $\sigma^2_{\text{dep}}/(Rm)$ without adding roots.
5. Multiplicity Boundary	Strict separation between the prespecified 24-estimate confirmatory family and post-outcome diagnostics (64-estimate family, proxy gap, break-even, root-deletion audits).

Limitations and Scope Boundaries

- **Single procedure and synthetic market:** All empirical evidence is conditional on one frozen replay-proxy update procedure and one controlled synthetic continuous double auction.
- **Saved-outcome reanalysis:** Current results are a reanalysis of saved outcomes rather than a full raw-model rerun.
- **No broad generalization claims:** The study makes no claim of a universal scaling law, general agent reliability, real-market execution validity, or isolated causal proxy bias.
- **Unmeasured operational dimensions:** Does not measure API monetary cost, token consumption, inference latency, safety constraints, or long-horizon reliability.
- **Auxiliary planner-code case:** The planner-code illustration uses only 3 generation roots and serves strictly as an auxiliary check, not as statistical evidence for unrestricted coding or autonomous agents.

Takeaways and Conclusion

1. **Evaluate the full frozen procedure:** Deployment utility must account for candidate screening, no-update decisions, and interactive feedback at the level of independent generation roots ($R = 50$).
2. **Scale dominates confirmatory conclusions:** Under the prespecified 24-estimate family for U_{100}, all four largest-scale cells ($s = 0.3$) are significantly negative (-11.98 to -22.01 bps), while smaller scales ($s = 0.002, 0.05$) and response contrasts remain unresolved.
3. **Proxy gains do not transfer directly:** At the selection scale ($s = 0.05$), post-outcome whole-root diagnostics show a positive gap of 0.952 to 1.314 bps between saved replay-proxy gains and interactive deployment gains.
4. **Completion preferences matter:** Post-outcome revaluation across $25, 50, 100$, and 200 bps along the affine frontier $\mu(p) = a + p\,b$ demonstrates how incomplete-fill penalties govern sign and magnitude at large order scales.
5. **Disciplined reporting:** Distinguishing prespecified joint confirmatory inference from post-outcome diagnostics prevents overclaiming from multi-condition evaluation suites.

Research blog

An a16z-style technical reading note

Complete research package
5 November 2026
Release 202611050855

Why Proxy Wins Don't Equal Deployment Value: Evaluating Policy-Update Procedures End to End

Research Blog

Every machine learning and decision-systems team eventually faces the same release question: *If we let our automated pipeline propose policy updates, screen them on an offline proxy, and ship the winner whenever it clears a threshold, does the deployed system actually improve over keeping the incumbent?*

In practice, teams routinely answer a different question. They inspect the proxy score of the selected candidate—or run a narrow post-hoc check on a single winning policy—and treat that number as the expected value of the update workflow.

That shortcut confuses the score of a selected candidate with the utility of the procedure that produced it. When an engineering organization puts a policy-update workflow into production, it is not deploying a single static policy in a vacuum. It is deploying a composite mechanism: candidate generation, offline screening, an adoption gate that either promotes a candidate or retains the incumbent, and execution in an interactive environment whose scale and feedback dynamics may differ from the screening proxy.

A new paper, *Evaluating the Deployment Utility of a Frozen Policy-Update Procedure: Scale, Completion Preference, and Joint Evidence*, isolates this distinction in a controlled applied study. Rather than proposing a new trading strategy or a universal agent benchmark, the study formulates and executes a bounded policy-evaluation protocol around a single estimand: the **expected incumbent-relative deployment utility of a frozen update-selection procedure**.

The findings offer a sharp cautionary lesson for ML engineering leaders, research managers, and risk owners. Even when an update procedure regularly finds candidates that look positive on a replay proxy, its expected utility in interactive deployment can remain statistically unresolved at small operational scales—and turn decisively negative when deployed at larger scale.

The Deployment Question: Candidate Scores vs. Procedure Utility

Consider how modern ML and quantitative engineering teams iterate on decision policies. Whether tuning execution algorithms, routing logic, or automated bidding rules, the workflow usually follows a standard template:

1. **Generate** a batch of candidate policy modifications from an existing baseline.
2. **Screen** those candidates against logged or replayed historical trajectories using a fast proxy metric.
3. **Select** the top-performing candidate if its proxy improvement exceeds a safety margin; otherwise, **retain the incumbent** (a no-update outcome with zero relative gain).
4. **Deploy** the resulting choice into the live, interactive environment.

When a candidate clears the bar with a +1.3 basis-point (bps) gain on the screening proxy, it is tempting to log +1.3 bps as the expected contribution of the update cycle. Yet three distinct forces drive a wedge between that proxy readout and actual deployment utility:

- **Selection optimism (winner's curse):** Choosing the best of K noisy candidates

systematically inflates the apparent quality of the winner, even if the proxy is unbiased for any single fixed policy.

- **Finite proxy-sample noise:** Screening on finite replay windows introduces estimation error into both candidate ranking and threshold gating.
- **Replay-to-interactive divergence:** A non-reactive replay log cannot capture how an environment responds when a modified policy changes order sizes, timing, or inventory trajectories.

Because of these compounding effects, evaluating the *procedure* requires treating the entire generate-screen-gate-deploy loop as a single frozen object of study. If the procedure decides not to update on a given run, its incumbent-relative utility is zero. If it adopts a candidate, its utility is whatever that candidate actually achieves in interactive execution relative to the incumbent, evaluated under the operator's true completion and cost preferences.

An Illustrative Case: Updating Execution Policies in a Synthetic Market

To make this concrete without confounding from non-stationary live markets, the study examines a frozen replay-proxy policy-update procedure operating in a controlled synthetic continuous double auction (CDA).

In this environment, an execution policy must liquidate or acquire a target inventory over a fixed horizon inside an order book populated by background market participants. The engineering team has an incumbent policy and a frozen update procedure that generates candidate parameter updates, evaluates them on replay data, and either adopts the best candidate or retains the incumbent.

What happens when that selected policy is placed into the interactive market?

At tiny order sizes, the policy's orders barely disturb the book; replay assumptions roughly hold, though run-to-run market noise is high relative to the tiny signal. As the order scale increases, however, the policy's own limit and market orders consume available liquidity, shift queue priority, and leave larger unexecuted remainders at the end of the horizon. If the operator cares deeply about completing the full order—penalizing incomplete fills heavily—a candidate policy that looked thrifty on passive replay data can suffer severe end-of-horizon completion penalties in interactive trading.

Crucially, this failure mode is invisible if you only examine the replay proxy score, and it cannot be diagnosed merely by counting how often a proxy-selected candidate beats the incumbent.

Methodology: Defining the Estimand and Unit of Analysis

The paper's primary contribution is methodological clarity: aligning the statistical protocol with the actual operational unit of a policy-update pipeline.

The Estimand and Experimental Structure

The target estimand is the expected incumbent-relative utility of the frozen update-selection procedure across a specified grid of deployment conditions. In the main synthetic-market study, the experimental design is structured around:

- **50 independent generation roots:** Each root represents an independent run of the candidate-generation process.

- **16 candidates per root:** Within each root, the procedure generates 16 candidate updates for screening.
- **100 root–regime selection decisions:** Each of the 50 independent roots produces one selection decision in each liquidity regime. The independent sample size is 50 roots, not 100 decisions.
- **2,400 paired interactive deployment records:** Selected policies and incumbents are evaluated across paired interactive market rollouts spanning a 12-cell deployment grid.

That 12-cell grid crosses three operational dimensions:

1. **Liquidity regime:** `liquid` vs. `thin` order books.
2. **Order scale (s):** small ($s = 0.002$), medium ($s = 0.05$), and large ($s = 0.3$).
3. **Inventory-response setting (ρ):** $\rho = 0$ vs. $\rho = 1$.

Together with key contrasts across regimes and responses, the original confirmatory design prespecifies a **24-estimate family** evaluated under an original utility definition U_{100}, which applies a **100-basis-point penalty** to incomplete fills at the end of the trading horizon.

Why the Root Is the Independent Unit

A common pitfall in ML systems evaluation is treating every evaluation rollout—here, all 2,400 paired deployment records—as independent observations. Doing so dramatically understates uncertainty.

When a single generation root produces a candidate pool and a selection decision, all interactive rollouts downstream of that root share the same candidate draw and screening noise. The independent experimental unit is therefore the **root** ($n = 50$), not the individual rollout. All valid confidence bands and simultaneous intervals in the study respect this root-level clustering.

Three Elementary Formal Characterizations

To guide how teams interpret deployment data, the paper lays out three elementary mathematical characterizations. These are not deep new theorems or novel statistical estimators; rather, they are straightforward accounting identities that prevent common analytical mistakes:

1. **Reversal frequency cannot determine mean utility:** Teams often track "sign reversals"—how frequently a candidate with a positive proxy score delivers negative utility in deployment. Elementary expectation algebra shows that reversal frequency alone does not determine whether the procedure has positive or negative expected utility. A procedure with a low reversal rate can still be net-negative if its occasional failures are severe (tail losses from incomplete fills), and a high-reversal procedure can be net-positive if its wins outweigh small losses. Gain and loss magnitudes must be retained.
2. **Linear preference frontier under fixed selections:** When deployment utility trades off execution price against an incomplete-fill penalty p (keeping the procedure's adoption choices fixed), expected utility is an affine function of the penalty parameter: $\mu(p) = a + pb$, where a captures the base execution component and b captures the expected completion shortfall relative to the incumbent. This yields a simple linear preference frontier and a closed-form diagnostic break-even penalty without re-running simulations.
3. **Two-level variance decomposition:** Total uncertainty in estimated deployment utility decomposes cleanly into *between-root variation* (differences across candidate genera-

tion and selection draws) and *within-root deployment noise* (interactive rollout variance for a given selection). Knowing which component dominates tells engineering teams whether they need more independent generation roots or more interactive rollouts per root.

Where This Fits in the Literature

To keep its scope sharp, the paper explicitly separates procedure-level deployment evaluation from several adjacent research areas:

- **Off-Policy Evaluation (OPE) and High-Confidence OPE:** Standard OPE methods estimate the value of a *fixed* target policy using logged data from a behavior policy. They do not evaluate an end-to-end procedure that generates random candidate pools, screens them, and deploys them across shifting interactive scales.
- **Safe Policy Improvement (e.g., SPIBB):** Safe improvement algorithms construct policies guaranteed to beat a baseline under specific support assumptions. They are *methods for building policies*, whereas this study audits the empirical deployment utility of an already-frozen update workflow.
- **Conservative Evaluation of Offline Policy Learning:** A close precedent in *Transactions on Machine Learning Research* (TMLR) studies conservative evaluation of offline learning pipelines. The present work shares this commitment to rigorous evaluation of learning workflows, extending the focus to interactive scale sensitivity, completion-penalty frontiers, and joint simultaneous inference across deployment conditions.
- **Active Offline Policy Selection:** Active selection methods adaptively allocate a limited budget of online evaluations to pick the best candidate from a set. Here, the selection rule is already frozen and offline; the question is what happens *after* selection when the procedure's output meets an interactive environment.
- **Agent Reliability and Benchmark Stability (e.g., HAL):** Recent benchmark audits examine ranking stability, scaffold sensitivity, and evaluation variance across generalist AI agents. While complementary in spirit, this study is a bounded applied policy-evaluation analysis, not a general-purpose agent leaderboard.
- **Transition-Level Fidelity (PIVOT and PIVOT-KG):** Prior work on PIVOT investigates transition-level *Improvement Fidelity* and knowledge-guided mechanisms (PIVOT-KG). The present paper evaluates frozen procedure-level deployment utility on its own standalone protocol and neither replaces those methods nor reuses PIVOT experimental results.

The Evidence: What Happened When the Frozen Procedure Met the Market

When the frozen update procedure is evaluated across the prespecified 24-estimate family under the original U_{100} utility metric (100 bps incomplete-fill penalty) using an approximate simultaneous confidence band, a striking pattern emerges across order scales.

1. Largest Scale ($s = 0.3$): Decisively Negative Utility

At the largest order scale ($s = 0.3$), the procedure's expected incumbent-relative utility is unambiguously negative across every liquidity regime and inventory-response setting. None of the simultaneous intervals cross zero:

Regime	Inventory Response (ρ)	Mean Utility U_{100} (bps) [Simultaneous Interval]
liquid	0	-11.9800 [-18.3944, -5.5656]
liquid	1	-12.2848 [-18.6992, -5.8704]
thin	0	-22.0073 [-28.4217, -15.5929]
thin	1	-21.6446 [-28.0590, -15.2302]

In liquid books at $s = 0.3$, deploying the frozen update procedure loses roughly 12.0 to 12.3 bps relative to simply keeping the incumbent policy. In thin books, the degradation nearly doubles, losing 21.6 to 22.0 bps.

The same saved point estimates are interpreted more cautiously in the wider 64-estimate diagnostic family: at 100 bps, the two liquid-regime largest-scale bands include zero, while the thin-regime estimates remain negative. The four negative largest-scale conclusions above belong to the prespecified 24-estimate confirmatory family; the wider 64-estimate result is a post-outcome sensitivity analysis.

2. Lower Scales ($s = 0.002$ and $s = 0.05$) and Contrasts: Unresolved

What about smaller order scales, or differences between inventory-response settings ($\rho = 0$ vs. $\rho = 1$) and interaction contrasts? Under the prespecified 24-estimate family, **all lower-scale cells and response/interaction contrasts remain statistically unresolved**. Their simultaneous intervals straddle zero.

This split outcome—decisively negative at large scale, unresolved at small and medium scales—is a central empirical takeaway. A team testing only at small scales might conclude the update procedure is harmless or mildly promising; a team extrapolating proxy wins to large orders would suffer immediate, statistically significant losses.

3. Post-Outcome Diagnostic: The Proxy-vs.-Deployment Gap at $s = 0.05$

Why did the procedure adopt updates if they did not reliably improve interactive performance? To inspect the mechanism, the authors conduct a separate outcome-aware, post-outcome diagnostic outside the prespecified 24-estimate family at the intermediate scale $s = 0.05$, comparing the saved replay-proxy gain (where a no-update decision is scored as 0) against the paired interactive deployment gain.

Evaluated with a separate whole-root simultaneous interval (outside the original 24-estimate family and without running new simulations), the gap between saved proxy gain and interactive deployment utility at $s = 0.05$ is consistently positive across all four cells:

- **liquid, $\rho = 0$:** 1.314 bps $[0.649, 1.979]$
- **liquid, $\rho = 1$:** 1.294 bps $[0.629, 1.959]$
- **thin, $\rho = 0$:** 0.952 bps $[0.288, 1.617]$
- **thin, $\rho = 1$:** 1.057 bps $[0.392, 1.722]$

In every cell, the replay proxy overstates interactive deployment performance by roughly 1.0 to 1.3 bps.

It is vital to state what this diagnostic is—and what it is not. It is **not** an isolate of causal proxy bias. Because it compares the selected candidate's saved proxy gain (with no-update $= 0$) to its interactive realization, it jointly reflects three entangled factors: winner-selection

optimism across the 16 candidates, finite proxy-sample noise, and structural differences between non-reactive replay and interactive order-book execution.

4. Post-Outcome Sensitivity: Completion Preferences and Break-Even Frontiers

How much of the large-scale failure is driven by the operator's 100-bps penalty on incomplete fills?

To answer this without pretending to run a new confirmatory experiment, the study performs a post-outcome utility sensitivity analysis. Keeping the exact same candidate selections and saved trajectories fixed—**without reselecting candidates** under new objectives—the analysis revalues the outcomes at incomplete-fill penalties of **25, 50, 100, and 200 bps**.

Using the affine preference frontier $\mu(p) = a + pb$ alongside a separate joint component construction, the authors map how expected utility shifts as completion preference varies and compute diagnostic break-even penalties. The four estimated zero crossings range from 0.454 to 5.236 bps; these are post-outcome ratio estimates with a separate uncertainty construction, not a confirmatory threshold. Along with an expanded **64-estimate family**, root-deletion stability checks, and alternative confidence band constructions, these sensitivity tools are transparently labeled as **post-outcome diagnostics**. They illuminate *why* the procedure loses money at scale—and how sensitive conclusions are to operational preferences—without blurring the line between prespecified confirmatory claims and exploratory reanalysis.

Scope Conditions and Explicit Limits

Strong engineering research is defined as much by what it refuses to claim as by what it proves. Decision-makers reading this study should keep five explicit boundaries in mind:

1. **One frozen procedure, one synthetic market:** All primary results apply to a single frozen replay-proxy update procedure inside one controlled synthetic continuous double auction.
2. **No universal scaling laws or real-market execution guarantees:** The degradation at $s = 0.3$ is not a universal mathematical law of policy scaling, nor does it validate real-exchange market microstructure execution or isolate causal proxy bias.
3. **No operational systems overhead:** The utility metric captures execution and completion preferences within the market task; it does not measure API costs, token/compute budgets, inference latency, safety guardrails, or long-horizon multi-day drift.
4. **Auxiliary scope of the planner-code case:** An accompanying planner-code case included in the broader project has only **three generation roots** and is strictly auxiliary; it does not support population-level statistical inference.
5. **Saved-outcome reanalysis:** The sensitivity, break-even, and proxy-gap analyses are reanalyses of saved experimental outcomes, not full raw-model reruns under redesigned selection objectives.

Practical Implications: A Checklist for Deployment Reports

For ML engineering leads and research managers, the operational lesson is clear: **stop approving policy-update pipelines based on candidate-level proxy metrics alone.** When

an automated workflow generates, screens, and gates policy updates, the workflow itself must be audited as a statistical procedure across the conditions where it will actually run.

To translate this principle into engineering governance, teams should require a structured **Deployment Procedure Report** before promoting an automated update workflow. Rather than treating such a report as an automatic green light to deploy, leaders should use it as an auditable specification—and a blueprint for future evaluation tooling.

Every deployment report for a policy-update procedure should explicitly document the following seven elements:

1. **The Frozen Procedure Specification:** Define the exact candidate-generation mechanism, candidate pool size (K), screening dataset or replay window, and hyperparameter configurations locked prior to evaluation.
2. **The Adoption and No-Update Rule:** Specify the exact gating threshold or selection criterion under which a candidate replaces the incumbent, and confirm that no-update outcomes are scored as zero incumbent-relative gain in procedure-level averages.
3. **The Independent Population Unit:** Identify the true independent unit of randomness (e.g., the generation root or episode seed cluster, such as the 50 roots here) rather than inflating sample size by counting correlated downstream rollouts ($2,400$ records) as independent.
4. **The Target Deployment Conditions:** Enumerate the full grid of operational environments (e.g., liquidity regimes, traffic volumes, order scales s, and environmental feedback settings ρ) across which the procedure is expected to operate.
5. **The Utility Function and Operational Costs:** State the exact evaluation objective—including completion penalties (such as the 100-bps penalty in U_{100}) and any unmodeled compute, token, or latency budgets—and note whether sensitivity checks revalue fixed selections ($\mu(p) = a + pb$) or require full reselection.
6. **The Inferential Family and Multiplicity Control:** Clearly separate the prespecified confirmatory family (e.g., the 24-estimate family with simultaneous confidence bands) from post-outcome diagnostics (e.g., expanded 64-estimate grids, proxy-vs.-deployment gap decompositions, or break-even frontiers).
7. **Triaged Conclusions (Positive, Negative, Unresolved):** Report results transparently across the deployment grid by classifying every cell and contrast as *statistically positive*, *statistically negative* (like the $s = 0.3$ cells), or *unresolved* (like the lower-scale cells and response contrasts), resisting the urge to round unresolved intervals up to success.

By separating candidate proxy scores from procedure-level deployment utility—and enforcing strict boundaries between confirmatory inference and post-outcome diagnostics—engineering teams can catch scale-induced failures in the lab before they become expensive surprises in production.